

数值实验部分严格审稿与系统重写方案

针对 bare_jrnl_new_sample4(6).pdf 的 Section XI

检查目标：逻辑压缩 · 创新点强化 · 巡检背景统一 · 实验融合

审稿结论

当前实验结果数量充足，但“六个算例对应六个小节”的组织方式使论证碎片化，且若干实验重复回答相近问题。建议将六组实验压缩为四个研究问题，并把巡检应用贯穿实验设置、任务语义、结果解释和在线修复。整体建议为 Major Revision。

报告包含：问题诊断、融合结构、逐段修改、完整英文替换稿、图表重组和补实验清单

1. 总体评价与最重要的审稿风险

当前 Section XI 位于论文第 13-16 页，包含 warm-start BnB、自动机构造消融、首解质量、机器人规模、基线比较和机器人失效修复六个小节。数据覆盖面较广，但章节没有形成“研究问题 - 对照设置 - 证据 - 结论”的闭环。严格审稿人会认为：实验多，但核心创新的证据被分散，应用背景停留在少量名词替换，部分结果甚至没有被完整讨论。

实验问题过度拆分

A 与 C 都在说明“首解是否好、BnB 是否能继续改进”；D 与 E 都在说明“随机器人规模增加是否仍有优势”。单独成节造成重复。

创新证据未按机制组织

论文的核心机制是 GBA 层剪枝、NBA/plan graph 同步构造、warm-start BnB 和失效后重规划，但当前标题以一般性能指标为主，读者需要自行推断各实验对应哪个创新。

巡检背景没有贯穿

实验设置仍以抽象 a_i 、 l_i 和 robot type 表示，结果解释主要谈“assignment nodes”和“automaton burden”，没有说明这些量为何对应大型设施巡检中的启动时延、任务连续性和故障恢复。

统计和复现信息不足

随机初始位置的实验多数给出单次四位小数结果。只有 First-Solution Quality 声称运行 25 次，但表中没有均值、标准差或置信区间。

选择性讨论结果

Table III 只讨论 first solution 与 plan-graph best 的差距，却没有讨论 plan-graph best 与 global optimum 的 9.35%-21.97% 差距；Table V 中 MILP 在 ϕ_1 上获得更低代价，也没有解释。

图表表达不规范

Fig. 2 的上下界曲线密集且振荡，无法对应标准 BnB 的全局 incumbent 和 frontier lower bound；Fig. 3 应称 Gantt chart，且字体、图例和失败后的重分配不够清楚。

2. 推荐的实验融合结构：六组压缩为四组

建议保留“实验设置”作为独立开头，然后用四个研究问题组织全部数值结果。这样既不丢数据，又能减少一个完整页面左右的重复说明，并让每个核心机制都有明确证据。

原小节	推荐新位置	合并理由
A. Warm-Started BnB	新 B: Anytime Performance and First-	与原 C 都回答首解质量和后续改进，Fig. 2

原小节	推荐新位置	合并理由
	Solution Quality	作为代表性过程，Table III 作为多任务汇总。
B. Planning-Aware Automaton Construction	新 A: Ablation of GBA Pruning and On-the-Fly Allocation	保留为独立核心消融，但将 Tables I-II 合并为一个表。
C. First-Solution Quality	新 B: Anytime Performance and First-Solution Quality	与原 A 合并，避免重复解释 first/best solution。
D. Scalability	新 C: Scalability and Comparison with Existing Methods	与原 E 都改变机器人规模。Table IV 可转为曲线或移至补充材料。
E. Comparison With Baselines	新 C: Scalability and Comparison with Existing Methods	主文保留代表性规模和 speedup，完整大表放补充材料。
F. Robot Unavailability	新 D: Online Repair after Robot Unavailability	独立保留，用于证明在线适应能力；增加“从头重规划”对照。

最终四个研究问题

RQ1: GBA-level pruning 与 on-the-fly allocation 分别减少了什么开销？

RQ2: 首个可执行巡检计划有多快、多好，warm-start BnB 能继续改善多少？

RQ3: 当巡检机器人数量和任务复杂度增加时，首解时延和最终代价如何变化，且与现有方法相比如何？

RQ4: 巡检机器人失效后，是否能从当前执行状态快速修复未完成任务？

3. 将实验统一为大型设施巡检背景

目前正文第 5 页已经出现 warehouse/factory 语义，但实验仍然像纯算法 benchmark。建议不改变任何数据，只统一任务解释。避免为每个 a_i 编造完全不同的业务故事，而应给出一个稳定的抽象映射。

抽象元素	推荐巡检语义	写作原则
Workspace W	40 m x 40 m 的工业设施巡检区域；障碍物表示设备、货架和禁止进入的通道。	只在实验设置中定义一次，后文直接引用。
Regions $l_1, \dots, l_8$	入口、仓储区、生产线、电气柜区、危险品区、维护间、物流通道和应急/充电区等巡检区域。	区域用途不必逐项重复。
Robot types	不同巡检能力，例如可见光、红外、气体检测、操作维护和空中巡检。	用“e.g.”表示示例，避免与未来实物平台冲突。
Action proposition a_i	在指定区域完成一次需要 n_i 台特定类型机器人的协同巡检、复核或维护操作。	统一为 inspection/maintenance operation。
T_{first}	从任务下达到得到首个可执行巡检计划的计算时延。	强调紧急检查和任务启动。
Robot unavailability	电量耗尽、传感器故障、通信中断或机械故障导致机器人退出。	与在线巡检连续性直接关联。

Recommended background paragraph:

The numerical experiments emulate inspection and maintenance in a large industrial facility. The eight regions of

interest represent operational areas that require different sensing or maintenance capabilities, while the obstacles represent machinery, storage racks, and restricted corridors. Each robot type denotes a distinct inspection capability, and an action proposition a_i specifies a collaborative operation at region l_i requiring n_i robots of type τ_i . This interpretation is used throughout all experiments; the formulas differ only in task order, concurrency, and safety constraints.

4. 实验开头与评价协议的问题及修改

4.1 当前问题

- 开头用六项长句罗列实验目的，信息密度过高，且每一项都使用 “we evaluate / we study / we compare” 的重复句式。
- “All experiments are conducted ... with obstacles randomly generated” 与后面 “initial positions randomly generated” 没有说明随机化是否固定种子、是否对所有方法共享。
- 未说明 T_{search} 的停止条件。Table IV 和 V 使用 infinity，不符合论文表格规范，也无法复现。
- 未说明是否把 LTL-to-automaton translation、distance preprocessing、MILP model construction 和 BnB certification 都计入运行时间。
- 随机实验中的四位小数会给出虚假的精度感。

4.2 推荐修改

- 用四个 research questions 替代六项枚举。
- 所有随机初始位置实验使用相同种子，并在各方法间配对；建议至少 20-25 次独立运行。
- 报告 mean $\pm$ standard deviation，并给出 95% confidence interval；偏态明显时改用 median [IQR]。
- 把 infinity 改为 TO，并在正文明确 time limit 和 memory limit 的实际数值。
- 明确 timing boundary：从 LTL 公式提交开始，到首个完整可执行计划/最优证书产生为止；预计算距离是否包含必须说明。

5. 各实验小节的严格审稿意见

5.1 新 A：GBA 剪枝和 on-the-fly allocation 的消融

原 Section XI-B 是最直接支撑论文创新的实验，应放在最前。但当前只说 “planning-aware automaton construction”，该名称过于宽泛。建议直接以两个机制命名，避免自造术语。

标题建议

Ablation of GBA-Level Pruning and On-the-Fly Allocation

- 公式族 $\Phi_m = \bigwedge_{i=1}^m \text{eventually } a_i$ 适合控制 automaton size，但只覆盖独立 eventually obligations，不能单独代表 “complex LTL specifications”。应说明它是 controlled stress test。
- Variant I 和 Variant II 的差别应在一张表中清楚表达，避免读者在 Tables I-II 之间来回对照。
- 建议合并表格列为：m、Proposed T_{first} 、without on-the-fly allocation T_{first} 、without early GBA pruning T_{first} 、generated NBA transitions、transition reduction。

- $m=15$ 时 Proposed 比 Variant I 缩短 30.23%，比 Variant II 缩短 41.43%，且减少 57.78% 的被构造 transition。用百分比比只重复绝对时间更能突出机制。
- 原文 “Table ??”、多余右括号、双层括号 ((32768,737280)) 和 “when ($m=15$)” 必须修正。

5.2 新 B: Anytime 性能和首解质量

原 A 和 C 应合并。Fig. 2 给出一个代表性搜索过程，Table III 给出多个任务的首解、plan-graph best 和 reference optimum，从“过程”和“总体质量”两个层面回答同一个问题。

- 代表性算例中 J_{first} 从 47.21 改善到 43.32，改进 8.24%；122,362/138,145 的剪枝率为 88.58%。这些数据应集中在一段。
- Fig. 2 当前红色/绿色曲线密集并振荡。若图中是每个 popped node 的 LB/UB，不能称为全局 lower/upper bound。应绘制 incumbent upper bound $J_{\text{inc}}(t)$ 和 open queue 的最小 lower bound $L_{\text{min}}(t)$ 。
- Table III 不能只强调 first-to-plan-graph gap。plan-graph best 相对 reference optimum 在 Case 1、2、4 分别存在 21.97%、9.35% 和 15.23% 差距，必须主动解释：方法优先首解时延和在线可用性，不保证保留原问题的全局最优解。
- “each case is executed 25 times” 与表中的单个数值矛盾。需说明是 mean、median、best 还是 representative run，并报告离散程度。
- 四个完整公式和 11 个 $c(a_i)$ 占据大量版面。建议把公式和巡检语义放入一张 compact mission table，正文只解释 temporal features。

5.3 新 C: Scalability 与基线比较

原 D 和 E 高度重叠。Table IV 单独改变单类型机器人数量，Table V 又同时改变任务和 team size。对于主文篇幅，建议把 Table IV 转为一张曲线，或移到补充材料；主文用 Table V/曲线完成规模与基线的联合论证。

- 当前最大队伍均达到约 100 台机器人，因此两组实验可以共享“规模”主线。
- 建议主文画 T_{first} vs. total number of robots，对 Proposed、Poset 和 MILP 使用相同横轴；完整原始表放 supplement。
- 在 ϕ_2 , 5x20 中，Proposed 的首解分别比 Poset 和 MILP 快 78.4x 和 58.3x；在 ϕ_4 , 5x20 中分别快 152.1x 和 272.2x。应报告 speedup，而不是逐行复述表格。
- Table V 中 ϕ_1 的 MILP best cost 为 32.85，而 Proposed/Poset 为 51.48，MILP 低 36.2%（相对于 Proposed）或 Proposed 高 56.7%（相对于 MILP）。这必须解释，否则“final cost comparison”显得选择性。
- 若不同方法的 feasible set、低层路径模型或 objective 不完全一致，应先统一；不能直接把不可比的 J_{best} 放在同一列。
- 所有方法必须使用同一初始位置、同一 distance matrix、同一运行预算和同一 timing boundary。

5.4 新 D: 巡检机器人失效后的在线修复

该实验是最能体现实际应用性的部分，但当前仅展示了方法自身的时间线，缺少对照，因此创新证据不足。

- 增加 retained-plan-graph repair 与 full replanning from scratch 的对照；这不需要新增实验场景，只需对现有两次失败事件各运行两种重规划方式。
- 说明计时从 failure event 被检测并更新 robot set 后开始，是否包含故障检测和通信延迟。

- 第二次故障的首解时间 1.652 ms 明显小于第一次的 45.738 ms。应解释为剩余任务更少或当前节点更接近接受状态，而不是称第二次“more severe”后直接给出更快结果。
- “tasks such as p1, p5, and p6” 符号错误：p_i 在前文是 region proposition，任务应写 a_i。
- 统一使用 robot，不要在同一段混用 agent/robot。
- Fig. 3 改称 Gantt chart；标出 failure times、failed robots、reassigned actions 和 replanning intervals，并增大字体。

6. 推荐的图表压缩方案

当前图表	处理建议	修改后作用
Fig. 2 BnB bounds	保留但重画为 wall-clock time 下的 incumbent 和 frontier minimum lower bound；去掉密集节点曲线。	证明 warm start 和 anytime improvement。
Tables I-II	合并为一个 ablation table。	同时量化 on-the-fly allocation 与 early GBA pruning。
Table III	保留并新增 Gap_first、Gap_graph 两列，或在正文明确报告。	区分首解排序损失与 plan-graph 限制损失。
Table IV	移至 supplement，或转为 T_first/T_best 随 n 的折线图。	避免与 Table V 重复。
Table V	主文改为曲线或只保留代表性规模；完整表移补充材料。	突出 speedup，不让大表淹没结论。
Fig. 3 Gantt	保留并重画；增加 failure/replanning/reassignment 标记。	连接巡检连续性和在线修复。

7. 可直接替换的 Section XI 英文版本

使用说明

下面版本按“实验设置 + 四个研究问题”重写，保留当前已有数据，不虚构新的统计量。带方括号的项目必须用实际 time limit、重复实验统计或新增对照结果替换。

XI. NUMERICAL EXPERIMENTS

This section evaluates four questions. First, we isolate the effects of GBA-level pruning and on-the-fly task allocation during NBA construction. Second, we examine how quickly the framework obtains an executable plan and how much the warm-started branch-and-bound search improves that plan. Third, we evaluate scalability with respect to task complexity and team size and compare the method with poset- and MILP-based approaches. Finally, we evaluate online repair after robots become unavailable during mission execution.

A. Experimental Setup and Evaluation Protocol

The experiments emulate inspection and maintenance in a 40 m x 40 m industrial facility containing eight regions of interest. The obstacles represent machinery, storage racks, and

restricted corridors. Each robot type denotes a distinct inspection capability, such as visual, thermal, gas, manipulation, or aerial inspection. An action proposition a_i specifies an inspection or maintenance operation at region l_i that requires n_i robots of type τ_i . This interpretation is used throughout the experiments; the formulas differ in task order, concurrency, and safety constraints.

All methods were implemented in Cython 0.29.24 and executed on a workstation with a 3.40-GHz Intel Core i7-11390H processor and 16 GB RAM. Runtime is measured from submission of the LTL formula to generation of a complete executable plan. The time to the first feasible plan is denoted by T_{first} , the time at which the best incumbent is obtained by T_{best} , and the total search time by T_{search} . Runs that do not terminate within [insert the actual time limit] are reported as TO. For experiments with random robot initial positions, all methods use identical random seeds, and the results should be reported as [mean +/- standard deviation with 95% confidence intervals] over [number] runs.

B. Ablation of GBA-Level Pruning and On-the-Fly Allocation

We first isolate the two operations that distinguish the proposed construction procedure. The controlled specification family $\Phi_m = \text{conjunction from } i=1 \text{ to } m \text{ of eventually } a_i$ represents m inspection requests distributed across the facility. This family is used as a stress test because increasing m enlarges the automaton while leaving the robot requirements easy to interpret. Variant I retains GBA-level pruning but postpones task allocation until the pruned NBA has been completely generated. Variant II disables early GBA pruning and on-the-fly allocation; it generates the complete NBA and applies the same pruning operations afterward.

The results show that the two operations address different sources of delay. On-the-fly allocation reduces the waiting time before planning can begin, whereas GBA-level pruning avoids generating transitions that are later removed. At $m = 15$, the proposed method obtains the first plan in 723.6905 s, compared with 1037.2168 s for Variant I and 1235.5359 s for Variant II, corresponding to reductions of 30.23% and 41.43%, respectively. The proposed construction also reduces the number of generated NBA transitions from 737,280 to 311,296, a reduction of 57.78%. Therefore, the main benefit is not only a smaller final automaton; unnecessary transitions are prevented from entering the NBA and planning stages.

C. Anytime Performance and First-Solution Quality

We next evaluate the quality of the early executable plan and the subsequent improvement obtained by branch-and-bound. In the representative inspection mission shown in Fig. 2, the on-the-fly plan graph provides an initial plan with cost $J_{\text{first}} = 47.21$. This incumbent enables pruning after 0.2910 s. The search prunes 122,362 of 138,145 visited assignment nodes (88.58%) and improves the cost to $J_{\text{best}} = 43.32$ at 2.3679 s, an 8.24% reduction relative to the first plan. Fig. 2 should plot the incumbent cost and the minimum lower bound among open search nodes against wall-clock time; these quantities directly describe solution improvement and the remaining optimality gap within the retained plan graph.

Table [X] summarizes four inspection specifications containing sequential, concurrent, disjunctive, safety, and until constraints. The first solution equals the best solution in the generated plan graph in Cases 1 and 2. In Cases 3 and 4, its gaps are 1.83% and 4.47%, respectively. These results show that the ordering rule usually exposes a useful executable plan before the complete search is finished. However, the best solution in the retained plan graph differs from the reference optimum in Cases 1, 2, and 4 by 21.97%, 9.35%, and 15.23%, respectively. Thus, the experiment supports early feasible planning and subsequent improvement, but it does not establish global optimality for the original LTL-MRTA problem. This distinction should be kept consistent with the theoretical claims.

D. Scalability and Comparison with Existing Methods

We evaluate scalability by increasing the number of robots and the temporal structure of the inspection mission. The focused single-type experiment shows that T_{first} increases from 0.1427 s for five robots to 1.7447 s for 100 robots, whereas exhaustive search becomes substantially more expensive and is not completed for the largest teams within the prescribed limit. This difference agrees with the complexity analysis: greedy evaluation grows moderately with the number of compatible robots, while exhaustive assignment search combines robot choices across successive subtasks.

The proposed method is also compared with the poset-based method in [3] and the MILP-based method in [7] using the same formulas, robot initial states, distance data, objective, and stopping conditions. The main advantage is the latency of the first executable plan. For ϕ_2 with a 5 x 20 team, the proposed method obtains the first plan in 1.1497 s, which is 78.4 times faster than the poset-based method and 58.3 times faster than the MILP-based method. For the more complex ϕ_4 with a 5 x 20 team, the corresponding speedups are 152.1 and 272.2. These results support the claimed advantage when inspection must begin before full optimization is complete.

The final-cost results should be interpreted separately from first-solution latency. For several settings, all methods reach the same best cost, whereas for ϕ_1 the MILP solution has cost 32.85 compared with 51.48 for the proposed method. This outcome indicates that the generated plan graph can exclude lower-cost executions even though it permits a feasible plan to be found quickly. Accordingly, the proposed method should be presented as an anytime feasible planner with fast initialization and plan-graph optimization, rather than as a uniformly globally optimal solver.

E. Online Repair after Robot Unavailability

The final experiment evaluates continuity of an inspection mission after robots become unavailable. The LTL specification remains unchanged, but assignments involving the failed robots are invalidated. Robot 0 is removed at $t = 15$ s and Robot 20 at $t = 30$ s. After each event, the current automaton progress and the states of the remaining robots define the replanning root, and only the unfinished actions are reassigned.

The retained-graph procedure obtains repaired first plans in 45.738 ms and 1.652 ms for the two events and improves them in 69.032 ms and 33.599 ms, respectively. The shorter time after the

second failure should be explained by the smaller remaining mission or the position of the current state in the retained graph, rather than by the severity of the failure alone. The Gantt chart should identify the failed robots, the failure times, the replanning intervals, and the actions reassigned to surviving robots. No action after a failure should be assigned to an unavailable robot.

To isolate the benefit of retaining the generated plan graph, the same failure events should also be solved by reconstructing the automaton and plan graph from the current mission state. The comparison should report T_{first} , T_{best} , and final makespan for retained-graph repair and full replanning. This additional control directly supports the online-repair contribution without introducing a new scenario.

8. 推荐的表格与图注文本

8.1 合并后的消融表标题

TABLE I
ABLATION OF GBA-LEVEL PRUNING AND ON-THE-FLY ALLOCATION
Columns: m | Proposed T_{first} | Without On-the-Fly Allocation T_{first} | Without Early GBA Pruning T_{first} | NBA
Transitions Generated | Reduction (%)

8.2 Fig. 2 图注

Fig. 2. Anytime branch-and-bound performance for a representative inspection mission. The incumbent cost is nonincreasing as better feasible plans are found, while the minimum lower bound among open nodes indicates the best remaining certificate within the retained plan graph. The shaded interval, if added, denotes variability across repeated runs.

8.3 首解质量表标题

TABLE II
FIRST-SOLUTION QUALITY AND PLAN-GRAPH RESTRICTION
Columns: Case | J_{first} | J_{PG} | J_{ref} | Gap_{first} (%) | Gap_{graph} (%) | GBA Reduction (%) | Plan-Graph Reduction (%)

8.4 基线比较图注

Fig. 3. Time to the first feasible plan versus robot-team size for the proposed, poset-based, and MILP-based methods. All methods use identical task instances, initial robot states, travel-time data, and stopping conditions. Points marked TO exceed the prescribed time limit.

8.5 在线修复图注

Fig. 4. Gantt chart of the inspection mission under robot unavailability. Vertical lines mark failure detection, shaded intervals mark replanning, and arrows identify actions reassigned from failed robots to compatible surviving robots.

9. 必须补充或澄清的数据

- 所有随机实验的重复次数、随机种子和统计量。
- Table III 中 25 次运行后的 mean/median 定义以及 SD/95% CI。
- Table IV/V 的实际 timeout 和 memory limit; 把 infinity 改为 TO。
- “Global Optimum/Reference Optimum” 的获取方式、搜索空间和 objective。
- MILP、Poset 和 Proposed 是否包含相同的 automaton translation 和 preprocessing 时间。
- 在线修复计时是否包含 failure detection、state synchronization 和 graph re-rooting。
- 保留图修复 vs 从头重规划的对照。
- 实物实验章节目前为空；正式投稿前必须补充或删除相关承诺。

10. 术语、符号和语病的集中修改

当前写法	建议写法	原因
planning-aware automaton construction	GBA-level pruning and on-the-fly allocation	直接写机制，避免宽泛自造名词。
agent / robot 混用	robot	与问题建模和全文保持一致。
Gantt graph	Gantt chart	标准术语。
best solution	best incumbent / plan-graph optimum / reference optimum	区分不同搜索范围。
optimal value in Fig. 2	final certified plan-graph optimum 或 reference value	避免暗示原问题全局最优。
infinity in tables	TO	并在正文给出 time limit。
This is mainly because that	This occurs because	语法错误。
when (m=15)	when m = 15	删除多余括号。
Table ??	Table I	修复交叉引用。
tasks p1,p5,p6	actions a1,a5,a6	p_i 是区域命题，不是任务。
cardinalitie	cardinality / cardinalities	拼写错误。
5 x 10 denotes...	a 5 x 10 team contains five robot types and ten robots per type	表达紧凑。

11. 最终审稿判断

从结果数量看，论文不需要继续增加更多独立“算例组”。真正需要的是把已有结果重新组织成四个清晰的研究问题，并补齐统计、基线公平性和在线修复对照。按照上述结构修改后，实验部分可以从当前约 3.5 页压缩到约 2.5-3 页，同时更直接地支撑论文的核心机制。

优先顺序：P0 解释 Table III 的 plan-graph/global gap，并修复 Table V 的 objective/fairness；P1 合并 A+C、D+E，合并 Tables I-II 并重画 Fig. 2；P2 将巡检语义写入 setup 和在线修复，并补 retained graph 与 full replanning 对照；P3 补重复运行、SD/95% CI、timeout 和复现信息。